

1. (5 points) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that T is continuous.

2. (5 points) Find the following limits, or show that they do not exist:

(a) $\lim_{(x,y) \rightarrow (0,0)} \sin\left(\frac{x}{y}\right).$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}.$

Q1: Show $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, linear, is continuous.

→ Aka, to show, $\forall x \in \mathbb{R}^n, \forall \epsilon > 0, \exists \delta > 0$ such that
 $\|x - y\| < \delta, \forall y \in \mathbb{R}^n \Rightarrow \|f(x) - f(y)\| < \epsilon$, where $f = T$

Lemma: ~~XXXX~~

$\exists M \in \mathbb{R}, \forall h \in \mathbb{R}^n, \|T(h)\| < M \|h\|$

Pf: Consider $T(h)_i = \sum_{j=1}^n a_{ij} \cdot h_j$

~~and $T(h)$~~

Observe, via Cauchy Schwarz,

$$\sum_{j=1}^n a_{ij} \cdot h_j \leq \sqrt{\sum_{j=1}^n (a_{ij})^2} \cdot \sqrt{\sum_{j=1}^n (h_j)^2} = \|h\|$$

$$\therefore |T(h)_i| \leq \sqrt{\sum_{j=1}^n (a_{ij})^2} \|h\| \rightarrow M_i$$

→ Now, $\|T(h)\| = \sqrt{\sum_{i=1}^m (T(h)_i)^2} \leq \sqrt{\sum_{i=1}^m (M_i)^2} \|h\| = \sqrt{\sum_{i=1}^m M_i^2} \|h\|$
 let this be our M

$$\therefore \|T(h)\| \leq M \|h\|$$

→ Back to continuity:

Need δ so that $\|T(x) - T(y)\| < \epsilon$ ie, $\|T(x-y)\| < \epsilon$

so if we get $M \|h\| < \epsilon$, we are done, as $\|T(x-y)\| \leq M \|x-y\| < \epsilon$
 $h = (x-y)$

where M is the obtained term from Lemma 1

→ let $\boxed{\delta < \frac{\epsilon}{M}} \Rightarrow M \|x-y\| < M \frac{\epsilon}{M} = \epsilon$

ie $\|T(x-y)\| < \epsilon$, hence T is continuous.

Matrix of T :

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & & & \\ \vdots & & & \\ a_{m1} & & & a_{mn} \end{bmatrix}$$

$$Th = Ah = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \\ a_{m1} & & & a_{mn} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$